GLUCOSE PREDICTION · MACHINE LEARNING PROJECT

Glucose Twin

A virtual patient that sees a glucose spike coming, 2 hours ahead

CGM shows the present.
The twin shows what is likely next.

A patient-level model for glucose above 180 mg/dL in the next 2 hours, trained on the Shanghai T2DM CGM dataset, with reasons and what-if exploration in an interactive app.



2 h
warning horizon

15 min
CGM resolution

180
mg/dL spike threshold

0
patients shared across train / val / test

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Executive summary

The twin turns a CGM trace into a 2-hour early warning, tested on patients the model never saw



Problem

- Spikes above 180 mg/dL follow meals and trends
- CGM only displays the present
- Early notice allows earlier action, with clinician guidance

What we built

- Pipeline: clean, label, engineer features
- LightGBM spike classifier
- Streamlit twin with risk, reasons, what-if

How we tested

- Train / validation / test hold different patients
- Threshold tuned on validation only
- Compared with a glucose-now baseline

Honest limits

- Small, single-region dataset
- Notes on meals and insulin are noisy
- Research prototype, not a medical device

A CGM is a rear-view mirror; clinicians and patients need a windscreen

Readings arrive every 15 minutes, but the decision that matters is what happens in the next 2 hours

15 min between CGM readings. The trace shows only the past and now.

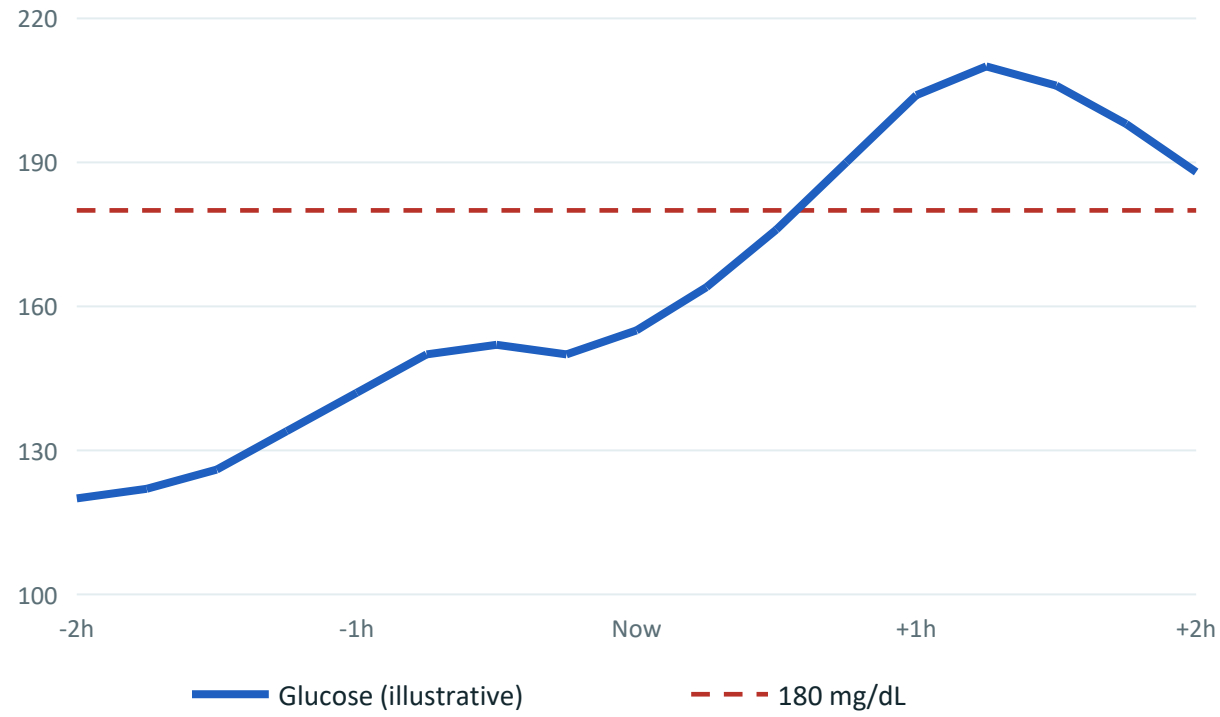
180 mg/dL: the threshold we call a spike.

2 h look-ahead: long enough to act, short enough to be predictable.

Goal

Turn the glucose trace into a spike probability, a high / low call and the reasons behind it.

What the twin looks at: past and now, then the next 2 hours



Illustrative curve, not patient data. A spike = any reading above 180 mg/dL in the next 8 readings.

Raw CGM and profile files become one clean, labelled table

The label looks only at the next 8 readings; nothing invented across long gaps

Sources

- Summary file: age, BMI, diabetes duration, HbA1c, flags
- Per-patient CGM files: timestamp and glucose
- Meal notes mark meal times
- Insulin dose columns, basal excluded
- T1DM supported with --kind T1DM

Cleaning rules

- Resample to a 15-minute grid
- Fill gaps of 30 min or less only
- Split recordings at longer gaps
- mmol/L converted to mg/dL if needed
- Drop values outside 40 to 500 mg/dL

The label




```
spike = 1  if  max(next 8) > 180
```

- The current reading is excluded, so the label cannot leak
- Rows near the end of a recording have no label and are not used for training

Patients	Readings	Spike rate	Test patients
[fill from train.py]	*[fill from train.py]*	*[fill from train.py]*	*[fill from train.py]*

Features describe level, trend, history, meals and the person

Every feature uses data up to the current reading, never later

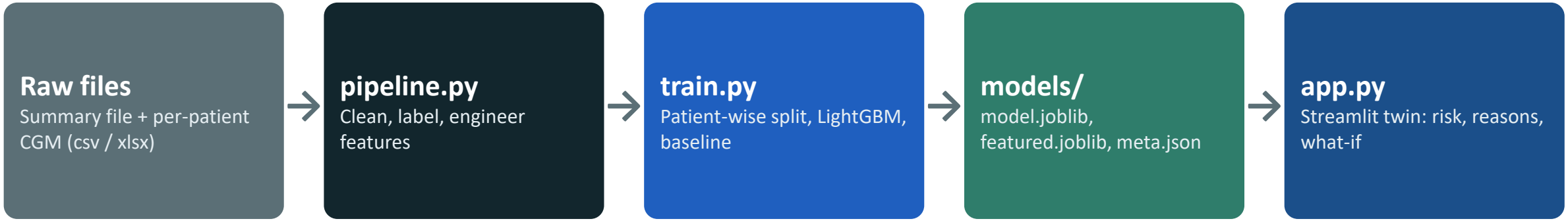
Group	Features	Why it helps
Current state	Glucose now	Starting point
Trend	Change over 15 min, 30 min, 1 h	Rising or falling fast
History	Mean, variability, peak, lowest over 1 h, 2 h, 4 h	Recent control and swings
Meals and insulin	Meals in last 2 h, hours since last meal (cap 6 h), insulin in last 2 h	Meals drive most spikes
Time of day	Sine and cosine of the hour	Daily rhythm, no midnight jump
Patient profile	Age, BMI, diabetes duration, HbA1c, one-hot flags from the Summary file	Baseline control per person

No-leakage rules

- Rolling windows look backward only
- Label uses the next 8 readings, never the current one
- Features are built per recording segment, so windows never bridge a long gap
- Hours since meal is capped at 6, so a missing meal note does not create extreme values
- Profile data is static per patient and joined by recording ID

Three scripts and a saved model make the pipeline reproducible

From raw files to an interactive twin in three commands



Run it

```
pip install -r requirements.txt
python train.py --data data
streamlit run app.py
```

Training takes about a minute and prints baseline vs model results. If LightGBM is missing, scikit-learn HistGradientBoosting is used instead.

Test without real data

- make_demo_data.py writes SYNTHETIC recordings in the same layout
- Use it only to check that the app runs
- Never report scores from demo data
- Real results come only from the Shanghai files

Evaluation is built so a good score means it works on new people

Different patients in every split, threshold set before the test, and a baseline to beat



Split by patient with GroupShuffleSplit: no patient appears in two sets.

Risk	Control in the code
Same patient in train and test	Group split by patient
Future information in features	Backward-only windows; label excludes the current reading
Threshold tuned on test data	Best-F1 threshold chosen on validation, then applied to test
Weak benchmark	Baseline: glucose now, with its own tuned threshold
Class imbalance	Class weighting in training; PR-AUC as headline metric

Metrics

- PR-AUC: headline, because spikes are the minority class
- ROC-AUC: how well risk is ranked
- Precision: how many alarms are real
- Recall: how many spikes we catch
- F1: balance, used to pick the threshold

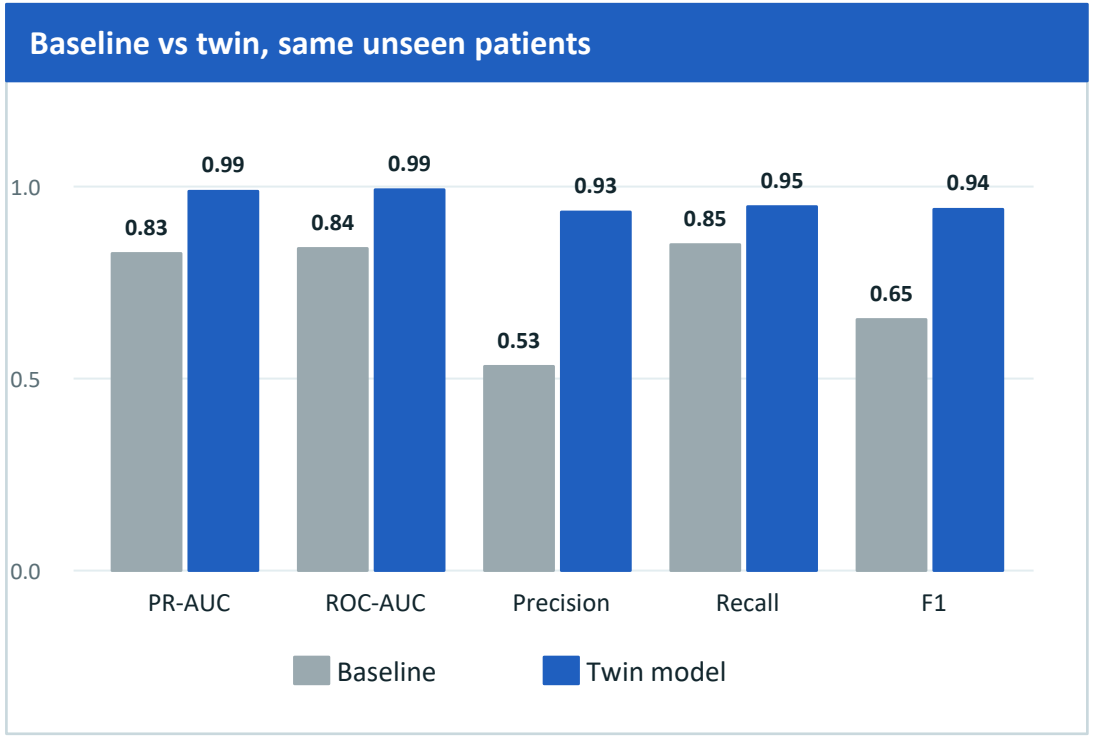
Results: the twin versus a glucose-now baseline on unseen patients

Run of train.py on the synthetic demo data (30 patients); swap in the Shanghai T2DM numbers before submitting

Metric	Baseline	Twin model	Change
PR-AUC	0.826	**0.988**	**+0.163**
ROC-AUC	0.839	**0.992**	**+0.153**
Precision	0.532	**0.934**	**+0.402**
Recall	0.849	**0.948**	**+0.099**
F1	0.654	**0.941**	**+0.287**

Baseline: predict a spike if glucose is already high, threshold tuned on validation. Test set: 6 unseen patients.

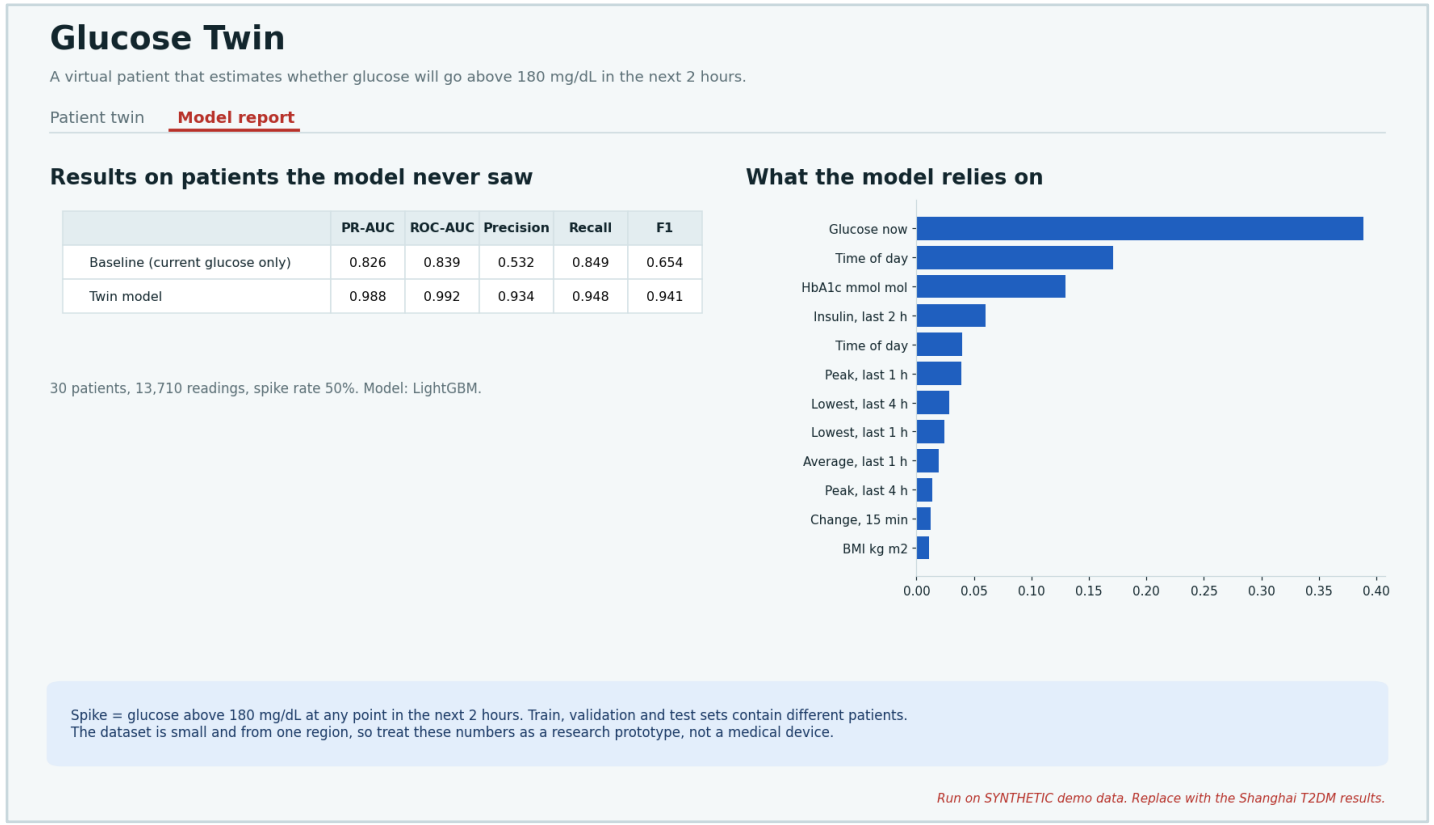
Demo-data run: synthetic, not real patients.



Takeaway: the twin beats the baseline by 0.163 PR-AUC (0.826 to 0.988) and 0.287 F1 (0.654 to 0.941) on patients it never saw, so trend, meal and history features add value beyond "glucose is high now".

The twin explains each prediction and lets a user test what-if changes

Pick a recording and a moment, see the risk, the reasons, and how the answer moves



1 **Choose a moment**
Recordings unseen in training are flagged

2 **Read the risk card**
Probability and a red or green verdict

3 **See why**
Top 7 factors: red raises risk, green lowers it

4 **Try what-if**
Glucose now, 1 h trend, hours since meal

What-if shows how the model responds, not a causal effect. Plain names such as "Hours since meal" replace raw feature names.

Seven design choices, each made to keep the evaluation honest

Choice, the easier alternative we rejected, and why

Decision	Easier alternative	Why we chose this
Split by patient	Random row split	Rows from one person are alike; random splits inflate scores
Threshold tuned on validation	Fixed 0.5 or tuned on test	Keeps the test set untouched
Baseline: glucose now	No comparison	Shows what trend, meals and history add
PR-AUC as headline	Accuracy	Spikes are the minority class; accuracy misleads
Fill gaps up to 30 min, split beyond	Interpolate everything	No invented values across long holes
Class weighting and early stopping	Plain fit	Handles imbalance and limits overfitting
Plain-language reasons	Raw feature names	Readable by non-technical users and clinicians

Limits and responsible use: a research prototype, not a medical device

Each limitation with its impact and what would close it

Limitation	Impact	Mitigation / next step
Small dataset, one region	May not generalise to other populations	External validation on other CGM datasets
Meal and insulin from patient notes	Missing or noisy context features	Cap on hours since meal; test without these features
Fixed 180 mg/dL threshold	Ignores personal targets	Personalised thresholds
What-if is model response	Not a causal effect	Label clearly; no treatment advice
Probabilities not calibrated	Percentages may over- or under-state risk	Calibration (Platt or isotonic) on validation
Not clinically tested	No safety evidence	Clinical review and prospective study first

Not for treatment decisions. Any real-world use needs clinical review.

Next: validate widely, personalise, then test clinically

What we would do next, and what the project already shows

Next (weeks)

- Add 30 min and 1 h horizons
- Predict low glucose too
- Probability calibration

Next (months)

- Per-patient personalisation
- Compare with LSTM / Transformer
- External dataset validation

Before real use

- Clinical review of outputs
- Prospective testing
- Regulatory and safety assessment

Conclusion

End to end
data to features to model to interactive app

Honest
patient-wise splits, a baseline, a tuned threshold

Explainable
a reason for every prediction, plus what-if

Thank you.

See the spike before it happens.

A 2-hour early warning for glucose above 180 mg/dL, built with past-only features, honest evaluation and plain-language reasons.

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